# BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SWAT

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2022

# PHYSICS

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

## SECTION-A

**Time Allowed:** 20 Minutes | **Maximum Marks:** 12

Note: 1) Attempting all MCQs is compulsory. This paper along with the OMR sheet must be returned to the superintendent after due time. 2) Fill the circle A B C D, which one is correct with blue or black ball point in separate OMR Sheet like ● 3) If more than one circle in the OMR sheet is filled then no credit will be given to such answer.

Q1. Slip rings are part of.

(A) DC motor
(B) AC generator
(C) Magnet
(D) Transformer

Q2. What is the number of neutrons in the plutonium $^{242}_{92}U$.

(A) 92
(B) 242
(C) 150
(D) 134

Q3. Electric (or) electromagnetic representations of data (or) information is termed as.

(A) Transmission
(B) Signal
(C) Software
(D) Computer

Q4. A step up transformer increases.

(A) Current
(B) Power
(C) Voltage
(D) Energy

Q5. When the current through a wire is 2.0A and voltage across its ends is 3.0V, then resistance.

(A) 1.5 Ω
(B) 2.5 Ω
(C) 6 Ω
(D) 4 Ω

Q6. The time rate at which the charge flows through the area is defined as.

(A) Voltage
(B) Electricity
(C) Current
(D) Magnet

Q7. The value of coulomb constant K depends on:

(A) Value of charges
(B) Material medium
(C) Separation of charges
(D) All of these

Q8. The overall magnification of a microscope is the product of the magnification produced by _____ lens.

(A) 2
(B) 3
(C) 5
(D) 6

Q9. A convex lense with focal length 8cm has the power of lens.

(A) 2.05 D
(B) 4.00 D
(C) 12.5 D
(D) 16.0 D

Q10. The speed of sound in dry air is 331 m/s and for each 1°C rise in temperature it increases by.

(A) 0.5 m/s
(B) 0.6 m/s
(C) 0.8 m/s
(D) 1 m/s

Q11. Which of the given affect the period of mass spring system.

(A) Mass
(B) Length
(C) Spring Constant
(D) Both mass & Spring constant

Q12. If the pendulum completes exactly 12 cycles in 2 min, what is the frequency of pendulum?

(A) 6 HZ
(B) 10 HZ
(C) 0.10 HZ
(D) 0.17 HZ

## SECTION "B"

**Time Allowed:** 2 hours and 40 minutes | **Maximum Marks:** 32

QII. Attempt any EIGHT Parts out of the following. Each Part carries equal marks.

(i) Why waves refract at the boundary of shallow and deep water?

(ii) What is the speed of sound in air at -20°C.

(iii) In which medium air (or) water an echo is heard sooner? Why?

(iv) Describe the term loudness, pitch and quality. Explain each by giving an example.

(v) Are rearview mirrors used in cars concave or convex?

(vi) The Sum of charges on both plates of a capacitor is zero. What does a capacitor store?

(vii) State and explain Ohm's law?

(viii) Differentiate between step-up and step-down transformers?

(ix) What is the function of an accelerating anode in an electron gun?

(x) How a flash drive is different from other storage devices?

(xi) If nuclear radiation is harmful. How it can be used for treatment diseases.

## SECTION "C"

**Maximum Marks:** 21

Note: Attempt any THREE questions of the following. Each question carries equal Marks.

QIII.

(a) [4] Using ripple tank explain reflection, refraction and diffraction of waves.

(b) [3] Water waves with wavelength 2.8m, produced in a ripple tank, travel with a speed of 3.80m/s. What is the frequency of the straight vibrator that produce them.

QIV.

(a) [4] What is total internal reflection? How we can calculate the critical angle for total internal reflection? What are the conditions for total internal reflection?

(b) [3] Explain the force on a current carrying wire in a magnetic field?

QV.

(a) [4] Describe the working principal of cathode Ray Oscilloscope (CRO) and make list of its uses.

(b) [3] What is email and internet? List few uses of internet in daily life.

QVI.

(a) [4] What are radioisotopes? Explain their three uses for various applications?

(b) [3] Iodine-131 is an important radioisotope for medical diagnostic and treatment procedures. The half life of $^{131}I$ is 8.02 days. Out of 100g of the sample how much will be left after 24 days.